

Compound Units: Speed, Density & Pressure

Quick reference sheet · GCSE Maths, Ratio, Proportion & Rates of Change · keep handy for revision

1. The three formulas

All three have the same shape: one quantity $\div$ another. Once you can rearrange one, you can rearrange all three.

Quantity	Formula	Units
Speed	speed = distance $\div$ time	m/s or km/h
Density	density = mass $\div$ volume	g/cm ³ or kg/m ³
Pressure	pressure = force $\div$ area	N/m ² = pascal (Pa)

2. Formula triangles

Cover the letter you want to find; the two remaining letters show the operation. Side-by-side (e.g. S next to T) means multiply; one on top of the other (e.g. D over T) means divide.

Speed: D over S $\times$ T $\rightarrow$ **S = D $\div$ T, D = S $\times$ T, T = D $\div$ S**

Density: m over $\rho \times V$ $\rightarrow$ **$\rho = m \div V$, $m = \rho \times V$, $V = m \div \rho$**

Pressure: F over P $\times$ A $\rightarrow$ **P = F $\div$ A, F = P $\times$ A, A = F $\div$ P**

3. Worked examples

Speed: distance 200 m, time 25 s $\rightarrow$ speed = $200 \div 25 = 8$ m/s

Density: mass 54 g, volume 20 cm³ $\rightarrow$ density = $54 \div 20 = 2.7$ g/cm³

Pressure: force 300 N, area 0.5 m² $\rightarrow$ pressure = $300 \div 0.5 = 600$ Pa

Rearranged: speed 30 m/s for 12 s $\rightarrow$ distance = speed $\times$ time = $30 \times 12 = 360$ m

For a fixed force, a **smaller area gives a bigger pressure** — e.g. a sharp knife or drawing pin.

4. Converting units

Conversion	Rule
km/h $\rightarrow$ m/s	$\div 3.6$
m/s $\rightarrow$ km/h	$\times 3.6$
g/cm ³ $\rightarrow$ kg/m ³	$\times 1000$
kg/m ³ $\rightarrow$ g/cm ³	$\div 1000$

$$\text{cm}^2 \rightarrow \text{m}^2 \qquad \div 10\,000$$

$$\text{m}^2 \rightarrow \text{cm}^2 \qquad \times 10\,000$$

$$72 \text{ km/h} \div 3.6 = \mathbf{20 \text{ m/s}} \quad \cdot \quad 2.7 \text{ g/cm}^3 \times 1000 = \mathbf{2700 \text{ kg/m}^3}$$

5. Common mistakes

Forgetting to convert units so they match before calculating (e.g. mixing minutes and hours, or cm and m).

Writing the answer with no units, or the wrong units (always check what the formula produces).

Multiplying instead of dividing (or vice-versa) when rearranging — use the formula triangle to check.